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Subject : DSA

Unit 2, S10, SLO1 - Doubly Linked List – Insertion at Beginning and End

Algorithm to insert a new node VAL at the beginning

Step 1: IF AVAIL = NULL

Write OVERFLOW

Go to Step 9

[END OF IF]

Step 2: SET NEW_NODE = AVAIL

Step 3: SET AVAIL = AVAIL -> NEXT

Step 4: SET NEW_NODE -> DATA = _____

Step 5: SET NEW_NODE -> PREV = _____

Step 6: SET NEW_NODE -> NEXT = _____

Step 7: SET _____ = NEW_NODE

Step 8: SET START = _____

Step 9: EXIT

Answer: Step 4: SET NEW_NODE -> DATA = VAL

Step 5: SET NEW_NODE -> PREV = NULL

Step 6: SET NEW_NODE -> NEXT = START

Step 7: SET START -> PREV = NEW_NODE

Step 8: SET START = NEW_NODE

Step 9: EXIT

Algorithm to insert a new node VAL at the end

Step 1: IF AVAIL = NULL

Write OVERFLOW

Go to Step 11

[END OF IF]

Step 2: SET NEW_NODE = AVAIL

Step 3: SET AVAIL = AVAIL -> NEXT

Step 4: SET _____ = VAL

Step 5: SET _____ = NULL

Step 6: SET PTR = START

Step 7: Repeat Step 8 while _____ != NULL

Step 8: SET PTR = PTR -> NEXT

[END OF LOOP]

Step 9: SET _____ = NEW_NODE

Step 10: SET _____ = PTR

Step 11: EXIT

Answer: Step 4: SET NEW_NODE -> DATA = VAL

Step 5: SET NEW_NODE -> NEXT = NULL

Step 6: SET PTR = START

Step 7: while PTR -> NEXT != NULL

Step 8: SET PTR = PTR -> NEXT

Step 9: SET PTR -> NEXT = NEW_NODE

Step 10: SET NEW_NODE -> PREV = PTR

Step 11: EXIT